The Sizland Land Acquisition Workflow

A Proprietary P2P/Escrow Interface

Bridging Blockchain Logic with Real-World Legal Milestones

System Architecture and Implementation Guide

Platform Features

- Multi-Signature Wallet Security
- Notary-Mediated Escrow System
- 4-Phase Workflow Implementation
- Document Hashing & Verification
- Fiat Bridge Integration (Crypto to KES)
- Smart Contract Automation

Document Version 1.0
January 23, 2026

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1 Executive Summary

The Sizland Land Acquisition Workflow is a comprehensive blockchain-based platform designed to facilitate secure land purchases through a multi-signature escrow system with notary mediation. The system eliminates technical barriers for rural landowners while providing transparency and security for international buyers.

1.1 Key Innovation

By keeping the seller off the platform, Sizland eliminates the technical barrier for rural landowners. The client only trusts Sizland, and Sizland uses its legal and blockchain rails to guarantee the deal.

1.2 Core Components

- **Smart Contracts**: MultiSigWallet.sol, NotaryMediatedEscrow.sol, EscrowFactory.sol
- **Frontend Interface**: React-based 4-phase workflow UI
- **Document System**: IPFS + on-chain hash verification
- **Fiat Bridge**: Cryptocurrency to Kenyan Shilling (KES) conversion
- **Notary Oracle**: Legal verification and milestone release authorization

2 The 4-Phase Workflow

The Sizland workflow is structured in four distinct phases, each with specific milestones and requirements.

2.1 Phase 1: Intake & Discovery

Objective: Capture client requirements and present curated land options.

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Phase 1 Activities

User Action:

- Client enters budget (e.g., $5,000)
- Specifies desired size (e.g., 2 acres)
- Indicates purpose (Farming/Speculation)

Sizland Action:

- Order routed to Sizland Admin Panel
- Team sources matching options
- Uploads Digital Gallery with:
  - Geo-pins (GPS coordinates)
  - Photos and videos
  - Property details and descriptions

Client Selection:

- Reviews available options
- Selects preferred plot
- Proceeds to due diligence phase

Smart Contract Function: `createIntake(uint _budget, uint _desiredSize, string memory _purpose, address _seller, address _notary)`

2.2 Phase 2: Due Diligence "Safety Gate"

Objective: Verify legal status and physical condition of selected property.

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Phase 2 Activities

Flexible Funding Options:

- Full escrow ($5,000), OR
- Initial deposit ($200)

Immediate Release:

- $200 released to Sizland Notary Address
- Covers registry search and site visit costs

Ground Truth Verification:

- Notary conducts registry search
- Physical site visit performed
- Comprehensive DD Report uploaded to client dashboard

Decision Gates:

1. **Proceed**: Client approves, moves to Phase 3
2. **Pivot**: DD reveals issues, client selects another plot
   - First 2 site visits covered by initial $200
   - 3rd visit requires top-up
3. **Cancel**: Client exits, remaining escrow refunded

Negotiated Rates:

- After successful Due Diligence, the Notary can set customized rates
- Negotiated Deposit Percentage (e.g., 5-20%)
- Negotiated Facilitation Fee (e.g., 1-5%)

Smart Contract Functions:

- `fundEscrow(uint _escrowId, bool _fullAmount)`
- `uploadDueDiligenceReport(uint _escrowId, bytes32 _reportHash)`
- `setNegotiatedRates(uint _escrowId, uint _depositBP, uint _facilitationBP)`
- `proceedWithPurchase(uint _escrowId)`

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2.3 Phase 3: Execution & The Fiat Bridge

Objective: Convert cryptocurrency to fiat and secure legal documents.

Phase 3 Activities

The Release:

- Upon client's "Proceed" command
- Purchase funds released from Escrow to Sizland Notary Address

The Fiat Bridge:

- Notary converts crypto to Kenyan Shillings (KES)
- Exchange rate recorded on-chain
- Seller paid offline in KES
- Transaction details logged in Admin Panel

Document Custody:

- Notary secures Original Title
- Obtains Signed Transfer documents
- Collects LCB (Land Control Board) Consents
- All documents placed in Sizland legal vault

UI Update:
Client sees: "Funds Disbursed. Sizland has secured all original documents in our legal vault."

Smart Contract Function: `executeFiatBridge(uint _escrowId, uint _kesAmount, uint _exchangeRate)`

2.4 Phase 4: Transfer Registry & Finalization

Objective: Complete land registry transfer and deliver title to client.

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Phase 4 Activities

The Registry Wait:

- Notary lodges transfer at Land Registry
- Typical duration: 2-3 weeks

UI Transparency:

- Official Lodgement Receipt uploaded
- Progress bar displayed
- Estimated Completion Date shown
- Regular updates provided

Title Issuance:

- New 99-year leasehold title issued in buyer's name
- Notary uploads digital scan to platform
- Original title prepared for shipping

Closing:

- Final facilitation fee released to Sizland Treasury
- Physical title shipped to client's European address
- Courier Tracking Number provided in UI
- Transaction marked complete

Smart Contract Functions:

- `uploadLodgementReceipt(uint _escrowId, bytes32 _receiptHash, uint _expectedComple`
- `completeTransfer(uint _escrowId, bytes32 _titleHash, string memory _courierTracki`

3 Smart Contract Architecture

3.1 MultiSigWallet.sol

Base multi-signature wallet implementation providing transaction security through multiple confirmation requirements.

Key Features:

- Configurable number of required signatures

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- Transaction submission and confirmation workflow
- Revocation capability
- Automatic execution upon reaching threshold

Core Functions:

```
function submitTransaction(address _to, uint _value, bytes memory
    _data)
function confirmTransaction(uint _txIndex)
function executeTransaction(uint _txIndex)
function revokeConfirmation(uint _txIndex)
```

Listing 1: MultiSigWallet Key Functions

3.2 NotaryMediatedEscrow.sol

Main escrow contract managing the entire 4-phase workflow with milestone-based fund releases.

Escrow States:

State	Description
Created	Initial escrow creation
Intake	Client requirements captured
Discovery	Digital gallery uploaded, awaiting selection
Funded	Escrow funded by client
DueDiligencePending	Awaiting DD report
DueDiligenceComplete	DD approved, ready for execution
ExecutionPending	Awaiting fiat bridge execution
ExecutionComplete	Funds converted and documents secured
TransferRegistry	Lodged at registry, awaiting title
TransferInProgress	FInalizing transfer
Finalizing	Title issued, preparing shipment
Completed	Transaction complete
Cancelled	Escrow cancelled

Milestone Structure:

1. **DueDiligence**: $200 for registry search and site visit
2. **Deposit**: Percentage-based deposit to seller
3. **FinalSettlement**: Balance + facilitation fee

Document Types:

- SearchCertificate
- DueDiligenceReport
- DraftSaleAgreement
- SignedSaleAgreement

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- SaleAgreement
- LodgementReceipt
- OriginalTitle
- LCBConsent

3.3 EscrowFactory.sol

Factory contract for modular escrow deployment, allowing multiple escrow instances with standardized configuration.

Features:

- Default Doddy notary configuration
- Custom notary support
- Batch escrow creation
- Centralized escrow tracking

Functions:

```
function createEscrowWithDoddy(address _seller, uint _totalAmount
    )
function createEscrow(address _seller, address _notary, uint
    _totalAmount)
function createBatchEscrows(address[] memory _sellers,
                            address[] memory _notaries,
                            uint[] memory _totalAmounts)
```

Listing 2: EscrowFactory Functions

4 Financial Flow

4.1 Fee Structure

For a typical $5,000 land purchase:

Component	Amount (Approx)	Customizable Percentage
Total Purchase Amount	$5,000	100% (Base)
Due Diligence Fee (Notary)	$200	Fixed ($200)
Deposit to Seller	$750	Negotiable (e.g., 15%)
Final Balance to Seller	$3,900	Remainder
Facilitation Fee (Sizland)	$150	Negotiable (e.g., 3%)

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4.2 Release Schedule

1. **$200** → Doddy (Due Diligence) - Released immediately upon funding
2. **$500** → Seller (Deposit) - Released after Sale Agreement signing
3. **$4,200** → Seller (Final Balance) - Released during fiat bridge
4. **$100** → Sizland Treasury (Facilitation Fee) - Released upon completion

4.3 Treasury Configuration

Sizland Treasury Address: `0x16de625c556118998e7bf74ed06c067f775771a1`
Multi-Sig Configuration:

- Doddy (Notary)
- Sizland Treasury
- Trusted Third Party
- Required Confirmations: 2 out of 3

5 Notary Admin Panel Features

Exclusive dashboard for Sizland team to manage escrows, document uploads, and client communications.

5.1 Dashboard

Escrow Management:

- Real-time status tracking for all active escrows
- Quick access to escrow details
- State visualization with color-coded badges
- Document security status indicators

5.2 Fiat Payment Logging

Capabilities:

- Record KES amounts paid to sellers
- Log exchange rates (Crypto to KES)
- Track crypto amounts released from escrow
- Maintain transparent payment history

Data Captured:

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- Escrow ID
- KES Amount Paid
- Exchange Rate (KES/USD)
- Crypto Amount Equivalent
- Timestamp

5.3 Timeline Management

Purpose: Keep clients informed during 2-3 week registry wait.

Features:

- Set Expected Completion Dates
- Update progress milestones
- Add notes for client communication
- Track lodgement receipt dates

5.4 Document Management Hub

Centralized location for all document uploads and verification.

Supported Documents:

- Search Certificate
- Due Diligence Report
- Sale Agreement (Draft and Signed)
- Lodgement Receipt
- Original Title (scan)
- LCB Consent

Document Workflow:

1. Upload document (PDF, JPG, PNG)
2. Generate SHA-256 hash
3. Store hash on-chain
4. Store original on IPFS (optional)
5. Mark as verified when appropriate

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6 Security Features

6.1 Multi-Signature Verification

- All fund releases require multiple confirmations
- Configurable confirmation threshold (default: 2 of 3)
- Transaction queue with pending/executed status
- Revocation capability before execution

6.2 Document Hashing & Verification

- SHA-256 hashing of all legal documents
- On-chain storage of document hashes
- Public verification capability
- Immutable audit trail
- IPFS storage for original documents

6.3 Access Control

- Role-based permissions (Client, Seller, Notary, Admin)
- Function-level modifiers enforcing authorization
- State-based access restrictions
- Notarycentric verification workflow

6.4 State Machine Protection

- Strict state transition enforcement
- Milestone completion validation
- Prevent out-of-order operations
- Emergency cancellation capability

7 Technical Implementation

7.1 Technology Stack

Smart Contracts:

- Solidity ^0.8.19
- Hardhat development environment

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- OpenZeppelin contract libraries
- Ethereum blockchain

Frontend:

- React.js
- ethers.js for Web3 integration
- MetaMask wallet connectivity
- Responsive UI design

Development Tools:

- Hardhat for compilation and testing
- Solhint for Solidity linting
- Slither for security analysis
- Mythril for symbolic execution

7.2 Deployment Process

1. Deploy MultiSigWallet with default owners
2. Deploy NotaryMediatedEscrow with treasury configuration
3. Deploy EscrowFactory with Doddy notary address
4. Verify deployment and configuration
5. Create test escrow for validation
6. Fund test escrow
7. Verify all contract functions

7.3 Network Configuration

Development:

- Network: Hardhat local node
- Chain ID: 1337
- RPC URL: http://127.0.0.1:8545

Production:

- Network: Ethereum Mainnet (or Polygon for lower fees)
- Proper gas optimization
- Multi-sig security fully activated
- Professional audit before deployment

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8 Operational Guidelines

8.1 Client Onboarding

1. Client connects MetaMask wallet
2. Enters purchase requirements (budget, size, purpose)
3. Reviews Digital Gallery of available plots
4. Selects preferred property
5. Funds escrow (full amount or initial $200)
6. Receives notification when DD report is ready

8.2 Notary Responsibilities

1. Receive due diligence fee ($200)
2. Conduct registry search
3. Perform physical site visit
4. Upload comprehensive DD report
5. Convert crypto to KES upon client approval
6. Secure all legal documents
7. Lodge transfer at Land Registry
8. Upload Lodgement Receipt
9. Update expected completion dates
10. Upload final title scan
11. Arrange courier shipping

8.3 Sizland Team Workflow

1. Monitor Admin Panel for new intake requests
2. Source suitable land options
3. Upload Digital Gallery (photos, videos, geo-pins)
4. Monitor escrow state transitions
5. Facilitate client-notary communication
6. Ensure document uploads are complete
7. Verify milestone completions
8. Process facilitation fee upon successful closure

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9 Compliance & Legal

9.1 Regulatory Considerations

- Compliance with Kenyan land transfer regulations
- Proper notary licensing and certification
- AML/KYC procedures for large transactions
- Cross-border payment regulations
- Cryptocurrency conversion compliance

9.2 Document Retention

- All document hashes stored on-chain permanently
- Original documents maintained in legal vault
- IPFS backups for redundancy
- 7-year retention policy minimum
- Audit trail for all transactions

9.3 Dispute Resolution

- Emergency escrow cancellation capability
- Multi-sig arbitration for disputes
- Legal recourse through Sizland entity
- Clear terms of service
- Client support infrastructure

10 Future Enhancements

10.1 Planned Features

- Mobile application (iOS and Android)
- Multiple currency support
- Automated KYC/AML integration
- Direct IPFS integration in UI
- Real-time chat with notary
- Email/SMS notifications

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- Multiple language support
- Bulk purchase discounts
- Referral program

10.2 Scalability Considerations

- Layer 2 scaling solutions (Polygon, Arbitrum)
- Gas optimization strategies
- Database indexing for faster queries
- CDN for document delivery
- Load balancing for high traffic

11 Conclusion

The Sizland Land Acquisition Workflow represents a novel approach to international land purchases, combining the transparency and security of blockchain technology with traditional legal processes. By serving as a trusted mediator and keeping sellers off-platform, Sizland removes technical barriers while providing clients with a secure, transparent path to land ownership.

Key Advantages:

- **Eliminates Technical Barriers**: Rural sellers don't need crypto knowledge
- **Trust Simplification**: Client only needs to trust Sizland
- **Hybrid Approach**: Blockchain security + traditional legal processes
- **Real Transparency**: Full UI visibility during registry wait
- **Professional Service**: Document custody, fiat conversion, shipping logistics
- **Legal Compliance**: Proper notarization and registry procedures

For technical support or implementation questions, contact the Sizland development team.

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